

- Q.26 Write a short note on mixed flow reactors.
- Q.27 Write in brief about Isobaric and Isothermal process
- Q.28 Differentiate between elementary and non-elementary reactions
- Q.29 Half-life period of a substance in a first order reaction is 20 minute. Calculate the rate constant
- Q.30 Draw the labeled sketch of plug flow and batch reactor.
- Q.31 Write a short note on Henry's Law
- Q.32 Define Vander Waal's equation and write its mechanical expression with the significance of its constant.
- Q.33 Explain closed and open system with one example each
- Q.34 Discuss about rate constant. Write the units of rate constant
- Q.35 Discuss any one of the following
i) Entropy ii) Free Energy

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 What are reactors? Explain the construction and working of any one type of reactor
- Q.37 Derive rate equation for the first order reaction
- Q.38 With the help of labeled diagram write all the steps involved in process of Carnot cycle and also write its efficiency.

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Roll No.

4th Sem / P & P

Subject:- Chemical Engineering Thermodynamics and Reaction Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a closed system?
- A sealed container of gas
 - An open bottle of water
 - The atmosphere
 - An insulated thermos with a lid
- Q.2 The first law of thermodynamics is a statement of
- Conservation of mass
 - Conservation of energy
 - Increase of entropy
 - Equilibrium
- Q.3 The efficiency of a Carnot cycle depends on:
- The nature of the working substance
 - The temperatures of the hot and cold reservoirs
 - The pressure of the system
 - The volume of the system
- Q.4 Activation energy of a reaction is:
- The energy required to start a reaction
 - The energy released during a reaction

- c) The difference in energy between reactants and products
 d) None of the above
- Q.5 Which of the following properties is intensive?
 a) Volume b) Mass
 c) Temperature d) Total energy
- Q.6 Raoult's law is applicable to:
 a) Ideal gas mixtures
 b) Non-ideal gas mixtures
 c) Ideal solutions
 d) Colloidal solutions
- Q.7 The first law of thermodynamics is primarily concerned with:
 a) Entropy b) Heat and work
 c) Temperature d) Pressure
- Q.8 During an isobaric process, the pressure of the system :
 a) Increases
 b) Decreases
 c) Remains constant
 d) Varies linearly with volume
- Q.9 For a zero-order reaction, the rate of reaction is:
 a) Dependent on the concentration of reactants
 b) Independent of the concentration of reactants
 c) Proportional to the square of the concentration of reactants
 d) Inversely proportional to the concentration of reactants

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- Q.10 In an adiabatic process, the heat exchange with the surroundings is:
 a) Maximum b) Zero
 c) Positive d) Negative

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What do you understand by the word enthalpy.
 Q.12 Define system
 Q.13 Define reactor
 Q.14 What is state function
 Q.15 Define activation energy
 Q.16 What is Raoult's law
 Q.17 State Zeroth law of thermodynamics
 Q.18 Write one example of molecularity of reaction.
 Q.19 What do you mean by equilibrium
 Q.20 Write the full form of PFR

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Differentiate between homogeneous and heterogeneous systems
 Q.22 Write ideal gas equation and calculate the value of universal gas constant R in $\text{m}^3 \text{atm/k.mol.k}$
 Q.23 Define second law of thermodynamic with its limitations.
 Q.24 Discuss Amagat's law and Dalton's law.
 Q.25 Write the effects of temperature and pressure on equilibrium constant.

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